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Lung Cancer Classification Using CNN and Random Forest

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***Abstract*—Lung cancer is one of the leading causes of cancer-related mortality worldwide. Early detection of lung cancer is crucial for improving survival rates and enabling more effective treatment options. However, manually detecting and classifying various forms of lung cancers is time-consuming and can be inaccurate. Traditional classification methods rely on manual feature extraction, necessitating the experimenter's expertise and professional knowledge. This research introduces a streamlined feature selection method using a deep convolutional neural network (CNN) to effectively tackle the issue. The aim of this proposed lightweight deep CNN is to enhance accuracy while reducing the number of parameters. The established approach involves three key stages: pre-processing, feature extraction, and classification. In the pre-processing phase, K-fold cross-validation ensures robust classification, complemented by data augmentation to augment the image dataset. In the feature extraction stage, modified CNNs effectively extract pertinent features, followed by Random Forest (RF) utilization to distinguish various forms of lung cancer from CT scan images. The proposed Deep CNN-RF model demonstrates effective detection and classification of various types of lung cancer, achieving an impressive accuracy rate of 97.1%.**

***Keywords*—Lung Cancer, Convolutional neural network, Deep CNN-RF, K-fold cross validation, Augmentation**

I. INTRODUCTION

Lung cancer is characterized by the fast and uncontrolled proliferation of malignant cells in the lung airways. Lung cells have the potential to become malignant, leading to lung cancer, the primary cause of cancer-related fatalities, accounting for approximately 27% of all cases [1]. This cancer ranks as the second most lethal worldwide, causing more deaths each year than breast, colon, and prostate cancers combined [2]. The two main types of lung cancer are non-small cell lung cancer (NSCLC) and small cell lung cancer (SCLC). Non-small cell lung cancer represents 80% to 85% of all lung cancer cases and includes three distinct types: adenocarcinoma, squamous cell carcinoma, and giant cell carcinoma. These subtypes are related in therapy and prognosis, but they originate from different types of lung cells. Roughly 10% to 15% of instances of lung cancer are SCLC patients [3]. Although the prognosis for lung cancer is dire, fatality rates can be successfully reduced by proper cancer type classification and early detection of the disease. The most advanced diagnostic method for differentiating between human cancer cells at this time is computed tomography (CT) scanning [4]. Lung CT scan images are essentially multiple-angle X-ray images of a specific organ that are used to compare and contrast abnormalities with healthy cells. The manual identification of various lung CT scan images is a laborious task that can also be prone to errors. Consequently, it is essential to create a computer-aided detection system for lung cancer.

Deep learning techniques have shown encouraging outcomes in identifying lung cancer through CT scan images [5]. A tailored deep-learning framework for lung cancer detection utilizing CT scans demonstrates exceptional performance in early-stage diagnosis, with the potential to transform lung cancer diagnostics [6]. This investigation employed deep learning techniques on a dataset comprising 1190 CT scan images sourced directly from the Kaggle IQ-OTH lung cancer collection, employing advanced preprocessing techniques. Comparative analysis with CNN, Resnet50, and InceptionV3 highlighted the effectiveness of these methods in early lung cancer detection and risk assessment [7]. Another study implements a lightweight CNN architecture for classifying lung cancer images using CT scans. It achieves comparable performance to larger models with fewer parameters, and integrates Grad-CAM for enhanced interpretability by visualizing influential regions in predictions [8]. The research [9] introduces a precise lung cancer identification framework using deep learning. It utilizes a GoogLeNet-based CNN with transfer learning and Median Intensity Projection (MIP) from 3D CT scans, achieving 81% accuracy on the LIDC-IDRI dataset.

Accurate classification of lung cancer types remains a significant challenge in medical imaging due to limited annotated datasets and the subtle visual similarities across various cancer types. Existing approaches relying on CNN models often struggle with generalizability and efficiency, particularly in distinguishing between complex tissue structures within lung cancer images. Moreover, traditional classifiers may not fully leverage the extracted features for optimal accuracy. This study aims to address these gaps by proposing a modified CNN architecture combined with a Random Forest (RF) classifier, enhancing both feature extraction and classification accuracy. The main contributions are:

- Designed a lightweight CNN model to improve lung cancer classification accuracy with fewer parameters.
- Applied K-fold cross-validation and data augmentation for robust model performance.
- Employed a CNN-Random Forest approach, achieving 97.1% accuracy in lung cancer detection.

The paper is organized as follows: Section I provides an introduction to the article, while Section II covers a summary of earlier studies. In-depth descriptions of the proposed system's approach are provided in Section III, while Section IV presents the analysis and results, along with a comparison study. The last section, Section V, offers a summary of the closing thoughts.

II. Related Works

This study [10] by Kasaudhan et al. investigates the transformative impact of AI on lung cancer diagnosis, emphasizing advancements in accuracy, efficiency, and early detection. It traces the historical progression of AI in this field, highlighting significant improvements in diagnostic techniques such as CT and CXR. The research underscores AI's role in enhancing diagnostic accuracy, reducing false positives, and supporting early-stage screening through multiparametric clustering. Additionally, it explores AI's potential in personalized medicine and broader applications, including enhancing diagnostics, treatment planning, and predicting lung cancer.

Singh et al. conduct a comprehensive review [11] of the efficacy of various machine learning ensemble techniques in diagnosing lung cancer, emphasizing algorithms such as SVM, XGBoost, LightGBM, AdaBoost, CatBoost, and Random Forest. The findings indicated that boosting techniques, specifically AdaBoost and XGBoost, attained the highest accuracy scores of 96.77% and 96.76% respectively. The findings indicate that machine learning has considerable promise for improving lung cancer diagnosis and lowering mortality rates.

Qurina Firdaus et al. created a system for detecting lung cancer using CT-scan images [12]. This system comprises four primary stages: image pre-processing, segmentation, feature extraction (including area, contrast, energy, entropy, and homogeneity), and the classification of cancer as either benign or malignant. The system reached an accuracy of 83.33% in diagnosing, showcasing the efficacy of the method in enhancing early detection and classification of lung cancer.

III. Proposed Method

To overcome traditional classification challenges, a CNN-RF hybrid model is proposed for identifying and classifying various types of lung cancers. The method outlined in Fig. 1 is structured into three distinct sections: preliminary processing, feature extraction, along with classification. During the pre-processing phase, image augmentation is utilized to expand the dataset, while K-fold cross-validation is applied for sub-setting and normalizing the images. In the feature extraction section, features are extracted automatically through the use of a modified Deep CNN. In the classification section, the RF classifier is employed to distinguish among various types of lung cancers, including lung adenocarcinoma, lung benign tissue, and lung squamous cell carcinoma.

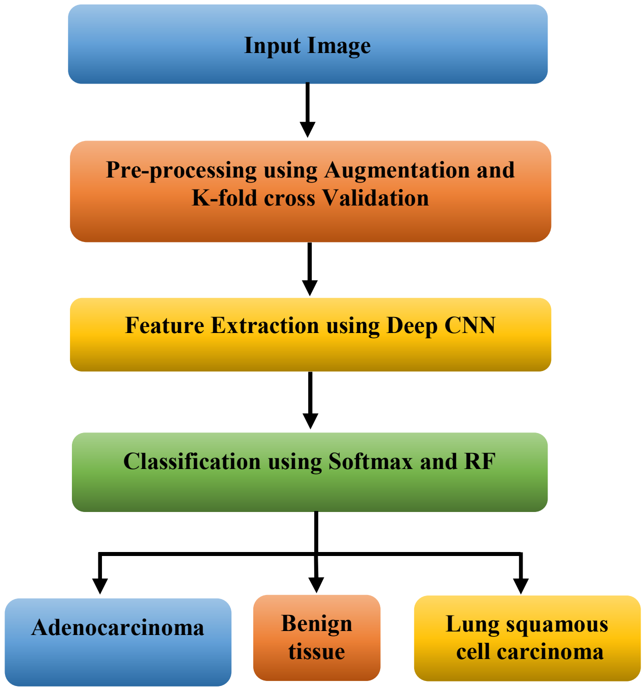
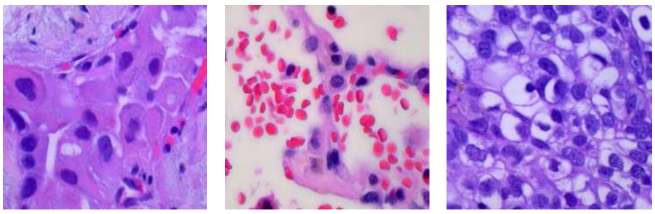


Fig. 1. Flow diagram of the proposed method

A. Dataset

This investigation utilizes a dataset obtained from Kaggle [15], which comprises histopathological images related to lung and colon cancer. The dataset comprises 25,000 JPEG images, each featuring a resolution of 768 x 768 pixels, categorized into five distinct groups. This study focuses exclusively on images related to lung cancer. The dataset contains images organized into three subfolders, with each subfolder containing 5,000 images: Lung benign tissue, Lung adenocarcinoma, and Lung squamous cell carcinoma. Fig. 2 presents sample images representing these three categories.



a. Lung adenocarcinoma | b. Lung benign tissue | c. Lung squamous cell carcinoma

Fig. 2. Types of lung cancer images

B. Image Pre-processing

The data is subjected to pre-processing prior to the training phase. The images are resized to dimensions of 60×60 pixels. The pre-processing step consists of two distinct phases: K-fold cross-validation and augmentation.

1) Data Augmentation

Image augmentation in machine learning is utilized to create new training samples and expand the dataset, thus preventing overfitting. This involves applying geometric transformations such as scaling, translation, and shearing along both the X and Y axes. These techniques help improve the model's generalization ability.

TABLE I. HYPER-PARAMETERS OF THE DEEP CNN MODEL

No	Deep CNN Model			
	Name	Type	Size of the activation	Learnable
01	Image input 60×60×1 images with 'Zero center'	Input Image	60×60×1	-
02	Conv_1 32 4×4×1 Convolutions with stride [1 1] and Padding is 'valid	Convolution	60×60×32	Weights 4×4×1×32 Bias 1×1×32
03	BN with 32 Channels	BN	60×60×32	Offset 1×1×32 Scale 1×1×32
04	Relu_1	ReLU	60×60×32	-
05	Maxpool_2 2×2 MP with stride [2 2]	MP	30×30×32	-
06	Conv_2 64 4×4×32 convolutions with the same stride and padding	Convolution	30×30×64	Weights 4×4×32×64 Bias 1×1×64
07	Relu_2	ReLU	30×30×64	-
08	Conv_3 84 3×3×64	Convolution	30×30×84	Weights 3×3×64×84 Bias 1×1×84
09	Batchnorm_2 BN with 84 channels	BN	30×30×84	Offset1×1×84 Scale 1×1×84
10	Relu_3	ReLU	30×30×84	-
11	Maxpool_2 2×2 MP	MP	30×30×84	-
12	Maxpool_3 2×2	MP	15×15×84	-
13	Conv_4 124 3×3×84	Convolution	7×7×84	Weights3×3×84×124 Bias 1×1×124
14	Batchnorm_3 BN with 124 channels	BN	7×7×124	Offset 1×1×124 Scale 1×1×124
15	Relu_4	ReLU	7×7×124	-
16	Maxpool_4 2×2 MP with stride [2 2]	MP	3×3×124	-
17	Conv_5 124 3×3×	Convolution	3×3×124	Weights 3×3×124×124 Bias 1×1×124
18	Batchnorm_4 BN with 124 channels	BN	3×3×124	Offset 1×1×124 Scale 1×1×124
19	Relu_5	ReLu	3×3×124	-
20	Fully Connected Layer_1	FC_1	1×1×1000	Weights 1000×1116 Bias 1000×1
21	Fully Connected Layer_2	FC_2	1×1×4	Weights 4×1116 Bias 4×1
22	Softmax	Softmax	1×1×4	-
23	Class output	Classification Output	-	-

2) K-fold Cross Validation

K-fold cross-validation involves dividing the dataset into K segments of equal size. The model undergoes training K times, with each segment utilized as the test set once, while the other segments contribute to the training process. This approach helps mitigate the impact of variability from any single train-test split, and the final performance is typically averaged over the K iterations.

C. Feature Extraction using Deep CNN

Using a typical Deep CNN architecture, lung cancer CT scans are automatically analyzed for features. Convolutional layers, batch normalization (BN), ReLU activation layers, and max pooling (MP) layers make up the Deep CNN architecture. The first convolutional layer uses a 4x4 kernel, padding, and 32 filters. To decrease overfitting, the feature map is reduced using max pooling, 2x2 stride, and ReLU activation layer after batch normalization. A 4x4 kernel size and identical padding ReLU activation layer follows the 64 filters in the next convolutional layer. Third convolutional layer includes 2x2 max pooling layers, ReLU activation layer, BN layer, and 84 3x3 filters. ReLU activation, max pooling, batch normalization, and 124 filters with a 3x3 kernel size comprise the fourth convolutional layer. 124 3x3 kernel filters make form the final convolutional layer after the BN and ReLU activation layers. The features from this last convolutional layer are coupled to two fully connected layers and classified using an RF classifier. The CNN-RF model classifies lung squamous cell cancer, lung benign tissue, and lung adenocarcinoma. Table I shows all suggested deep CNN model hyperparameters.

D. Classifier

A method known as Random Forest (RF) is employed for addressing both regression and classification challenges through an ensemble learning strategy. Throughout the training process, multiple decision trees are constructed, leading to the generation of the mean prediction for regression tasks or the mode of the classes for classification based on the individual trees. By averaging the outcomes from many decision trees, RF improves generalization and reduces overfitting while increasing prediction accuracy and robustness. Fig. 3 shows basic random forest architecture. The decision function for RF can be expressed as follows:

$$f(x) = \mathrm{mode}(\{T_1(x),T_2(x),\ldots,T_k(x)\}) \tag{1}$$

Where f(x) represents the final class label determined by the mode of predictions Ti(x) from k decision trees for the input x

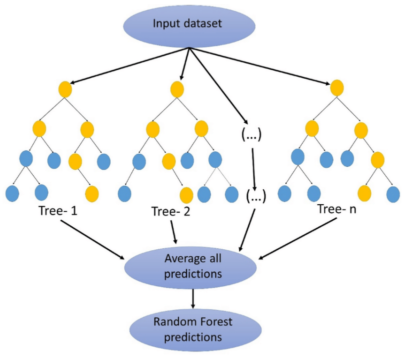


Fig. 3. Random Forest Architecture

IV. Result And Analysis

A modified deep CNN is employed for the detection and classification of lung cancers, utilizing both Softmax and RF classifiers. RF outperforms Softmax in terms of accuracy. The dataset is split into 80% for training and 20% for testing. The hybrid CNN-RF model achieves optimal accuracy with 18 epochs, a learning rate of 0.003, and the 'adam' optimizer. The proposed method is tested in a MATLAB R2023a environment in conjunction with a 12-core central processing unit and an NVIDIA RTX 3080 graphics processing unit.

A. Confusion Matrix

A confusion matrix is used to evaluate the classification performance of the proposed CNN-Softmax and CNN-RF hybrid models, as depicted in Fig. 4 and Fig. 5. In these matrices, the target classes are shown along the X-axis, while the output classes are displayed along the Y-axis.

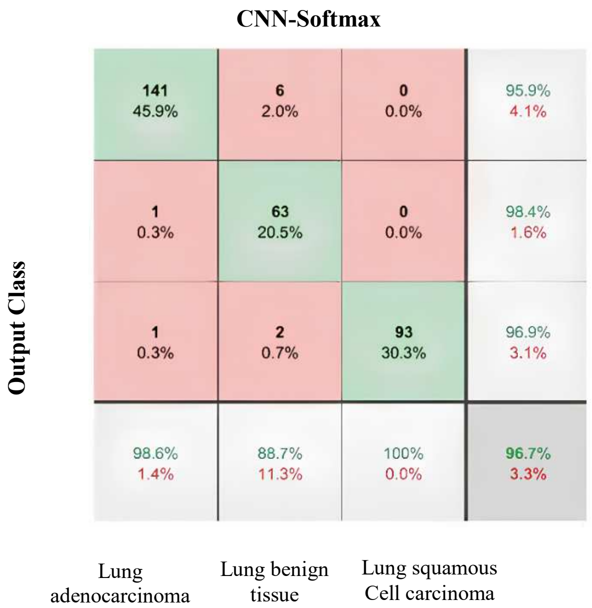


Fig. 4. Confusion matrix for the CNN-Softmax model

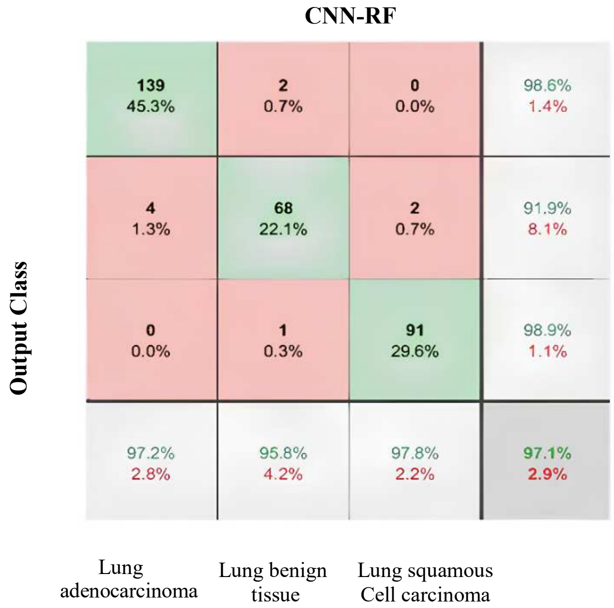


Fig. 5. Confusion matrix for the CNN-RF model

B. Performance Evaluation

This confusion matrix employs contributes of True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN) to facilitate the evaluation of metrics such as accuracy, specificity, sensitivity, and precision.

$$\text{Specificity} = \frac{TN}{TN+FP} \tag{2}$$

$$\text{Recall} = \frac{TP}{TP+FN} \tag{3}$$

$$\text{Precision} = \frac{TP}{TP+FP} \tag{4}$$

$$\text{F1 Score} = \frac{2\times \text{Precision}\times \text{Recall}}{\text{Precision}+\text{Recall}} \tag{5}$$

Using the equations provided, the sensitivity, precision, accuracy, and specificity for both the CNN+Softmax model and the CNN+RF model are presented in TABLE II and TABLE III, respectively.

TABLE II. CNN+SOFTMAX Model Performance

Lung Cancer Types	Precision	Specificity	Recall	F1 Score
Lung adenocarcinoma	98.6%	96.9%	95.9%	97.2%
Lung benign tissue	88.7%	99.6%	98.4%	93.3%
Lung squamous Cell carcinoma	100%	98.6%	96.9%	98.4%

TABLE III. CNN+RF Model Performance

Lung Cancer Types	Precision	Specificity	Recall	F1 Score
Lung adenocarcinoma	97.2%	98.8%	98.6%	97.9%
Lung benign tissue	95.8%	97.5%	91.9%	93.8%
Lung squamous Cell carcinoma	97.8%	99.5%	98.9%	99.5%

The model undergoes training and evaluation k times utilizing various subsets, such as 1, 3, 5, 7, and 9. The results of the performance analysis utilizing K-fold cross-validation are presented in Table IV. CNN-RF attains 92.9% accuracy in the first cross-validation, compared to CNN-Softmax's 92.7%. In the third cross-validation, CNN-RF attains 93.1% accuracy, whereas CNN-Softmax records 92.8%. CNN-Softmax achieves 93.4% and CNN-RF reaches 94.2% in the sixth cross-validation. CNN-RF receives 96.1%, whereas CNN-Softmax receives 95.5% for the sixth cross-validation. CNN-Softmax scores 96.7% in the eighth cross-validation, but CNN-RF reaches 97.1%. The ninth-fold cross-validation ultimately yields the best accuracy

TABLE IV. K-FOLD CROSS VALIDATION PERFORMANCE

K-fold	CNN+Softmax	CNN+RF
K= 1	92.7%	92.9%
K= 3	92.8%	93.1%
K= 5	93.4%	94.2%
K= 7	95.5%	96.1%
K= 9	96.7%	97.1%

The Fig. 6 presents the accuracy rates of several pre-trained models that were evaluated. Inception-V3, one of the early models, records an accuracy of 88.50%. ResNet-50, which includes residual connections for better performance, achieves 90.30%. VGG19, known for its deep architecture, shows a higher accuracy of 92.60%. Xception, reaches an accuracy of 94.90%. The proposed model, designed to enhance performance further, outperforms all others with an impressive accuracy of 97.10%.

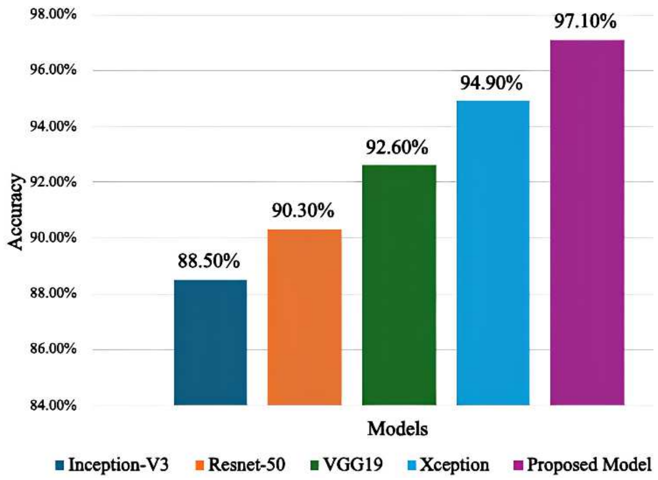


Fig. 6. Evaluation of various pre-trained models

The Table V, summarizes the accuracy of different models used in previous research for lung cancer detection. The GoogLeNet-MIP achieved 81%, GLCM-SVM attained 83.33%, and CNN-ALCD reached 94.11%. AdaBoost demonstrated a high accuracy of 96.77%. The proposed CNN-RF model surpasses all previous models, achieving an accuracy of 97.10%, indicating its superior performance in detecting lung cancer.

TABLE V. COMPARATIVE ANALYSIS

References	Model	Accuracy
[9]	GoogLeNet-MIP	81%
[12]	GLCM-SVM	83.33%
[14]	CNN	91%
[13]	CNN-ALCD	94.11%
[11]	AdaBoost	96.77%
Proposed Model	CNN-RF	97.10%

V. CONCLUSION

In conclusion, the present research aimed to establish a lightweight deep convolutional neural network (CNN) fused with Random Forest (RF) to enhance the accuracy of lung cancer identification and classification. The study addressed challenges associated with automatic feature extraction, which can be time-consuming and error-prone. The methodology was organized into three stages: pre-processing, feature extraction, and classification. K-fold cross-validation and data augmentation were applied to ensure robust classification, while modified CNNs were utilized for feature extraction. The Deep CNN-RF model achieved a high accuracy of 97.1%, showcasing its potential for diagnosing various types of lung cancer. This model represents a significant advancement in medical imaging applications, combining CNN and RF to enhance classification accuracy. Despite its effectiveness, the study is limited by the dataset size, which may affect generalizability, and potential overfitting concerns due to limited data variability. Future work will focus on validating the model with larger, more diverse datasets and exploring further optimizations to address overfitting and improve diagnostic accuracy.

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